Light

GCSE Computer Science — FDE Cycle and Systems

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GCSE • Edexcel • Grades 3—9 | Topic test | 30 min | 55 marks | Answered: 0 / 30

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Multiple-choice topic test for **GCSE Computer Science (Edexcel)**. You may skip questions. When you submit, you'll see your score, a grade estimate, and worked explanations. **Tip:** use the stopwatch above to time yourself.

Q1 (1 mark)

The CPU continuously repeats a cycle to process instructions.
What is the name of this cycle?

- ○ **A.** The fetch-decode-execute cycle
- ○ **B.** The input-process-output cycle
- ○ **C.** The read-write-store cycle
- ○ **D.** The start-run-stop cycle

Q2 (1 mark)

During the fetch stage of the FDE cycle, the CPU retrieves something from main memory.
What does the CPU fetch?

- ○ **A.** The next instruction to be processed
- ○ **B.** The result of a calculation
- ○ **C.** Data to be displayed on screen
- ○ **D.** A new program to install

Q3 (1 mark)

After fetching an instruction, the CPU must decode it.
What happens during the decode stage?

- ○ **A.** The instruction is encrypted for security
- ○ **B.** The instruction is converted from binary to text
- ○ **C.** The instruction is sent back to memory for storage
- ○ **D.** The Control Unit works out what the instruction means and what actions are needed

Q4 (1 mark)

The execute stage is the final stage of the FDE cycle.
What happens during the execute stage?

- ○ **A.** The instruction is deleted from memory
- ○ **B.** The instruction is displayed on the monitor
- ○ **C.** The instruction is copied to create a backup
- ○ **D.** The instruction is carried out by the appropriate CPU component

Q5 (1 mark)

The Program Counter (PC) is a register in the CPU.
What does the Program Counter store?

- ○ **A.** The address of the next instruction to be fetched
- ○ **B.** The result of the most recent calculation
- ○ **C.** The total number of programs installed on the computer
- ○ **D.** The current instruction being executed

Q6 (1 mark)

The Accumulator (ACC) is a register used during calculations.
What is the purpose of the Accumulator?

- ○ **A.** To store the memory address of data
- ○ **B.** To hold the current instruction being decoded
- ○ **C.** To store the results of calculations performed by the ALU
- ○ **D.** To count how many instructions have been executed

Q7 (1 mark)

The CPU communicates with memory using buses.
What is the purpose of the address bus?

- ○ **A.** To carry the memory address that the CPU wants to access
- ○ **B.** To carry data between the CPU and memory
- ○ **C.** To carry control signals like read and write

- ○ **D.** To supply electrical power to memory

Q8 (1 mark)

The data bus connects the CPU to main memory.
What does the data bus carry?

- ○ **A.** Only memory addresses
- ○ **B.** Only control signals
- ○ **C.** Data and instructions being transferred between CPU and memory
- ○ **D.** Only electrical power

Q9 (1 mark)

Many everyday devices contain embedded systems.
What is an embedded system?

- ○ **A.** A computer system built into another device to perform a dedicated function
- ○ **B.** A desktop computer that is hidden inside a desk
- ○ **C.** A laptop that has been permanently attached to a wall
- ○ **D.** A smartphone app that runs in the background

Q10 (1 mark)

Embedded systems are found in many household devices.
Which of the following contains an embedded system?

- ○ **A.** A paper notebook
- ○ **B.** A wooden chair
- ○ **C.** A microwave oven with digital controls
- ○ **D.** A manual wind-up clock

Q11 (1 mark)

Modern computers are based on the Von Neumann architecture. What is a key feature of the Von Neumann architecture?

- ○ **A.** Programs and data are stored in separate memory units that cannot communicate
- ○ **B.** Programs must be permanently wired into the CPU
- ○ **C.** Programs are stored on paper tape while data is in electronic memory
- ○ **D.** Programs and data are stored together in the same main memory

Q12 (1 mark)

The Memory Address Register (MAR) is used during memory access. What does the MAR hold?

- ○ **A.** The address of the memory location being accessed
- ○ **B.** The data being read from memory
- ○ **C.** The instruction currently being executed

○ **D.** The result of the most recent calculation

Q13 (2 marks)

During the fetch stage, several operations occur in sequence.
Which sequence correctly describes the fetch stage?

○ **A.** Instruction executed → Result stored → PC incremented

○ **B.** CIR decoded → Control signals sent → Operation completed

○ **C.** Data loaded into ACC → ALU performs calculation → Result stored in memory

○ **D.** PC copied to MAR → Address sent to memory → Instruction copied to MDR → Instruction copied to CIR → PC incremented

Q14 (2 marks)

The Memory Data Register (MDR) plays an important role in memory operations.
Which statement correctly describes the MDR?

○ **A.** The MDR only sends data from CPU to memory, never the other way

○ **B.** The MDR holds data being transferred between CPU and memory in either direction

○ **C.** The MDR stores the address of data, not the data itself

○ **D.** The MDR is only used during the execute stage, not during fetch

Q15 (2 marks)

The control bus carries signals that coordinate system operations.
Which signals are typically carried by the control bus?

- ○ **A.** Memory addresses and data values
- ○ **B.** Program instructions and calculation results
- ○ **C.** Read/write signals, clock pulses, and interrupt requests
- ○ **D.** Usernames, passwords, and encryption keys

Q16 (2 marks)

During the decode stage, the Control Unit processes the instruction in the CIR.
What does the Control Unit do during decode?

- ○ **A.** It performs the arithmetic calculation specified by the instruction
- ○ **B.** It stores the instruction result in the Accumulator
- ○ **C.** It interprets the instruction and prepares the control signals needed to execute it
- ○ **D.** It copies the instruction back to main memory for safekeeping

Q17 (2 marks)

The Current Instruction Register (CIR) holds information during instruction processing.
What is stored in the CIR and when is it used?

- ○ **A.** The CIR stores the memory address of the current instruction; used during fetch only
- ○ **B.** The CIR stores the next instruction to fetch; used during all three stages

- ○ **C.** The CIR stores calculation results; used during execute only
- ○ **D.** The CIR stores the current instruction being processed; used during decode and execute

Q18 (2 marks)

Embedded systems have specific characteristics that make them suitable for dedicated tasks.
Which characteristics are typical of embedded systems?

- ○ **A.** Large screens, full keyboards, and ability to run any software
- ○ **B.** Dedicated function, low power consumption, and often no user interface
- ○ **C.** High cost, large size, and require constant internet connection
- ○ **D.** General purpose, easily upgradeable, and run multiple programs simultaneously

Q19 (2 marks)

A CPU has a 16-bit address bus.
What does this tell us about the system?

- ○ **A.** The CPU can process 16 instructions per second
- ○ **B.** The CPU has 16 registers available
- ○ **C.** The CPU can transfer 16 bytes of data at once
- ○ **D.** The CPU can address up to 2^16 (65,536) different memory locations

Q20 (2 marks)

During the FDE cycle, the Program Counter is incremented.
When is the PC incremented and why?

- ○ **A.** After execute, to record how many instructions have been completed
- ○ **B.** After fetch, to point to the next instruction in sequence
- ○ **C.** Before fetch, to skip the current instruction
- ○ **D.** During decode, to calculate the instruction length

Q21 (2 marks)

A car contains multiple embedded systems.
Which components in a car would typically use embedded systems?

- ○ **A.** The windscreen glass, tyres, and seat fabric
- ○ **B.** The engine management system, anti-lock braking system (ABS), and airbag controller
- ○ **C.** The fuel tank, exhaust pipe, and door handles
- ○ **D.** The paint finish, floor mats, and sun visor

Q22 (2 marks)

The Von Neumann architecture has a limitation related to how it accesses memory.
What is this limitation?

- ○ **A.** It can only store programs up to 1 kilobyte in size

- ○ **B.** It cannot perform mathematical calculations
- ○ **C.** It cannot fetch an instruction and data at the same time because they share the same bus
- ○ **D.** It requires programs to be written in machine code only

Q23 (2 marks)

A computer has a 32-bit data bus.
What does this mean for data transfer?

- ○ **A.** The computer can only run 32-bit software
- ○ **B.** The computer can store 32 files at once
- ○ **C.** The computer has 32 gigabytes of RAM installed
- ○ **D.** The computer can transfer 32 bits (4 bytes) of data in a single operation

Q24 (3 marks)

An instruction 'ADD 200' means: add the value at memory address 200 to the Accumulator.
The ACC currently contains 50. Memory address 200 contains the value 30.
Which sequence correctly describes the execute stage for this instruction?

- ○ **A.** 200 is loaded into ACC; result is 200
- ○ **B.** Address 200 is placed in MAR; memory sends value 30 to MDR; ALU adds 30 to ACC (50); ACC now contains 80
- ○ **C.** Value 50 is written to address 200; ACC becomes 0

○ **D.** PC is set to 200; the next instruction is fetched from address 200

Q25 (3 marks)

Before a fetch operation, the registers contain:
- PC = 1000
- MAR = 500
- MDR = 42
- CIR = previous instruction

After the fetch stage completes, which registers will have changed?

○ **A.** Only PC will change (incremented to 1001)

○ **B.** Only MAR and MDR will change

○ **C.** PC, MAR, MDR, and CIR will all change

○ **D.** Only CIR will change to hold the new instruction

Q26 (3 marks)

A system has a 20-bit address bus.
What is the maximum amount of memory this system can address?

○ **A.** 20 bytes, because there are 20 address lines

○ **B.** 1,048,576 bytes (1 MiB), because 2^20 = 1,048,576 addressable locations

○ **C.** 20,000 bytes, because each bit represents 1000 bytes

○ **D.** 1,000,000 bytes (1 MB), because 20 bits rounds to 1 million

Q27 (3 marks)

A company is designing a smart thermostat for home heating control. Why is an embedded system more suitable than a general-purpose computer for this application?

- ○ **A.** An embedded system can display better graphics on the thermostat screen
- ○ **B.** An embedded system is lower cost, uses less power, is more reliable for continuous operation, and can be physically smaller
- ○ **C.** An embedded system allows users to install their own apps and games
- ○ **D.** An embedded system provides faster internet browsing speeds

Q28 (3 marks)

A simple program has these instructions in memory:
• Address 100: LOAD 500 (load value from address 500 into ACC)
• Address 101: ADD 501 (add value from address 501 to ACC)
• Address 102: STORE 502 (store ACC to address 502)
If the PC starts at 100, how many complete FDE cycles are needed to execute all three instructions?

- ○ **A.** 1 cycle, because the CPU processes all instructions together
- ○ **B.** 9 cycles, because fetch, decode, and execute are each counted as separate cycles
- ○ **C.** 6 cycles, because each instruction needs a fetch cycle and a separate execute cycle
- ○ **D.** 3 cycles, because each instruction requires its own fetch-decode-execute cycle

Q29 (3 marks)

A system with a 64-bit data bus needs to transfer 256 bytes from memory to the CPU.
How many transfer operations are required?

- ○ **A.** 256 transfers, because each byte requires its own transfer
- ○ **B.** 4 transfers, because 64 bits = 64 bytes per transfer
- ○ **C.** 32 transfers, because 64 bits = 8 bytes per transfer, and 256 ÷ 8 = 32
- ○ **D.** 1 transfer, because 64 bits can address 256 bytes

Q30 (3 marks)

A traffic light control system uses an embedded system rather than a desktop PC.
Which combination of factors best justifies this choice?

- ○ **A.** Traffic lights need to display videos and play music to entertain waiting pedestrians
- ○ **B.** Traffic lights need 24/7 reliability, real-time response to sensors, low maintenance, and weather resistance
- ○ **C.** Traffic lights need to connect to social media to report traffic conditions
- ○ **D.** Traffic lights need to run antivirus software and receive operating system updates

Total: **55** marks • Time: **30 min**

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